

Xiaoyue GAO

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EDUCATION

University of Wisconsin-Madison	WI, USA
Ph.D. Candidate in Pathology	09/2022-Present
Focus: B-cell lymphoma mechanisms, DNA methylation, and therapeutic resistance.	
Beijing Institute of Genomics, Chinese Academy of Sciences (CAS)	Beijing, China
Master of Science in Bioinformatics	09/2019-06/2022
Also affiliated with: China National Center for Bioinformatics (CNCB)	
University of Chinese Academy of Sciences	Beijing, China
Bachelor of Science in Biological Science	09/2015-06/2019
Australian National University	Canberra, Australia
International exchange	02/2018-06/2018

HONORS AND AWARDS

2025 ASH Graduate Hematology Award

- Awarded by the American Society of Hematology (ASH).
- Highly Competitive: Selected as 1 of only 12 recipients nationwide based on scientific merit.
- Funding: Awarded \$80,000 research grant to support innovative research in B-cell lymphoma.

2025 ASH Abstract Achievement Award

- Merit-based award honoring trainees with high-scoring abstracts at the ASH Annual Meeting.
- Recognized for excellence in scientific research presentation.

RESEARCH EXPERIENCE

Department of Medicine, University of Wisconsin-Madison	09/2022 – Present
<i>Doctoral Researcher</i> (Advisor: Prof. Lixin Rui)	
- Project: Elucidating the role of DNA methylation in mantle cell lymphoma (MCL) pathogenesis. - Designed and led independent research projects investigating epigenetic mechanisms of drug resistance. - Performed integrative multi-omics analysis (RNA-seq, ChIP-seq) and conducted in vivo validation using xenograft models.	
U.S. Food and Drug Administration (FDA) & UW-Madison	09/2023 – 03/2025
<i>Collaborator</i> (Advisors: Prof. Lixin Rui & Dr. Charles Wu)	
- Project: Analysis of Regulatory Botanical Submission Profiles for Cancer Management (FDA Perspective). - Led the biostatistical analysis of regulatory submission data to evaluate the landscape of botanical drug development. - Collaborated with FDA reviewers to synthesize findings into a regulatory review article (Published in <i>Therapeutic Innovation & Regulatory Science</i>).	
Beijing Institute of Genomics, Chinese Academy of Sciences (CAS)	09/2019 – 06/2022
<i>Graduate Researcher</i> (Advisor: Prof. Lan Jiang)	
- Project: Single-Cell Chromatin Accessibility Landscape of Human Medulloblastoma. - Conducted scATAC-seq data analysis to reconstruct the gene regulatory networks of brain tumors. - Characterized the tumor immune microenvironment and identified key transcription factors driving tumorigenesis.	

PUBLICATIONS

Peer-Reviewed Journal Articles

- **Gao X**, Zhuang Q, Li Y, Li G, Huang Z, Chen S, Sun S, Yang H, Jiang L, Mao Y. "Single-Cell Chromatin Accessibility Analysis Reveals Subgroup-Specific TF-NTR Regulatory Circuits in Medulloblastoma." *Advanced Science*. 2024 Aug;11(30):e2309554.
- Liu Y, Kimpara S, Hoang NM, Daenthanasanmak A, Li Y, Lu L, Ngo VN, Bates PD, Song L, **Gao X**, Bebel S, Chen M, Chen R, Zhang X, Selberg PE, Kenkre VP, Waldmann TA, Capitini CM, Rui L. "EGR1-mediated metabolic reprogramming to oxidative phosphorylation contributes to ibrutinib resistance in B-cell lymphoma." *Blood*. 2023 Nov 30;142(22):1879-1894.
- Park JK, Lee D, Rui L, **Gao X**, Furness MS, Wu C. "Analysis of Regulatory Botanical Submission Profile for Cancer Management from the U.S. FDA Perspectives." *Therapeutic Innovation & Regulatory Science*. 2025.
- Liu Y, Chen R, **Gao X**, et al. "PRMT5 inhibition sensitizes B-cell lymphoma cells to ferroptosis." *Leukemia*. 2026 April 17;40, pages1247-1260
- **Gao X**, Rui L. "DNA Methyltransferases in Hematologic Malignancies: Roles in Gene Regulation, Development and Disease Pathogenesis." *Genes & Diseases*. In press.

SELECTED PRESENTATIONS

Gao X, Hoang NM, Rui L. "DNMT3B functions as tumor suppressor in mantle cell lymphoma"

- Poster Presentation at *the 67th American Society of Hematology (ASH) Annual Meeting*, Orlando, FL. (2025).
 - Winner of ASH Abstract Achievement Award

Gao X, Hoang NM, Rui L. " DNMT3B functions as tumor suppressor in mantle cell lymphoma"

- Poster Presentation at *the University of Wisconsin-Madison Carbone Research Retreat*, Madison, WI. (2025).
 - Selected via competitive abstract review process.

TECHNICAL SKILLS

Bioinformatics & Computational Biology:

- **Programming & Environment:** Proficient in R (Bioconductor), Linux/Unix shell scripting.
- **NGS Data Analysis:** Expertise in processing and analyzing bulk and single cell multi-omics data (RNA-seq, ChIP-seq, ATAC-seq) to reconstruct regulatory networks.
- **Software & Tools:** SAMtools, FastQC, Bowtie2, MACS2, Seurat, DESeq2 and ArchR.
- **Biostatistics:** Applied statistical modeling for regulatory submission profiles (FDA collaboration).

Experimental Biology (Wet Lab):

- **Molecular & Cellular Techniques:** CRISPR/Cas9 gene editing, Western Blot, qPCR, Flow Cytometry/FACS, Immunofluorescence, Immunohistochemistry and Cell Culture (B-cell lines).
- **In Vivo Studies:** Murine xenograft models, In vivo drug efficacy testing, Tumor burden monitoring.